

Linear Function

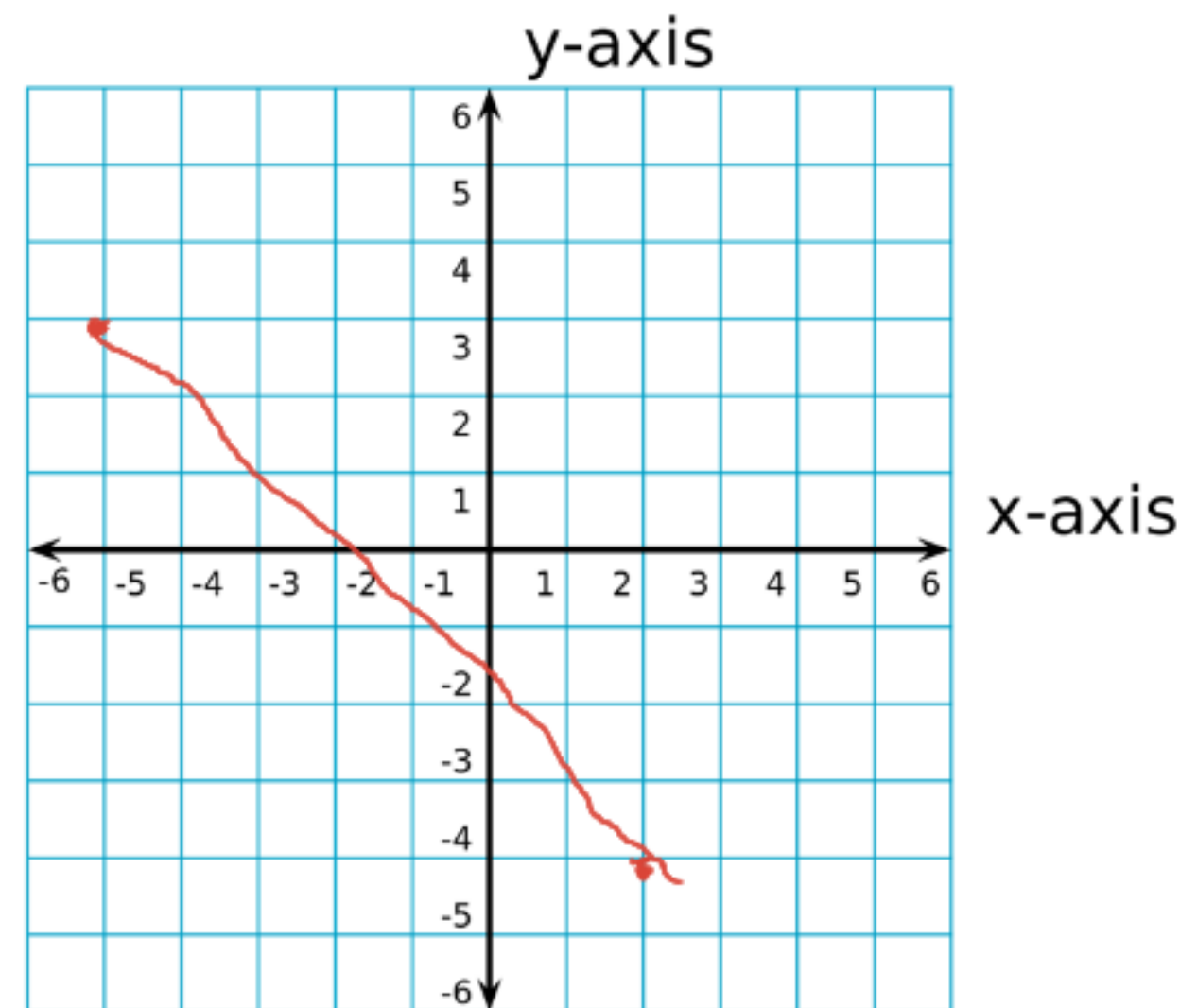
EXAMPLE

- 01 Draw a graph for the following functions:
 $f(2) = -4$ and $f(-5) = 3$

Rewrite it as ordered pairs

$$f(\underline{2}) = -4 \Rightarrow (2, -4)$$

$$f(-5) = 3 \Rightarrow (-5, 3)$$



Linear Function

EXAMPLE

02 Find the slope of a graph for the following function.
 $f(-3) = -1$ and $f(-8) = -6$

a) rewrite it as ordered pairs

$$f(-3) = -1 \Rightarrow (-3, -1)$$

$$f(-8) = -6 \Rightarrow (-8, -6)$$

b) use the slope formula to evaluate the slope

$$(-3, -1) \text{ and } (-8, -6)$$

$$(x_1, y_1) \quad (x_2, y_2)$$

c.) slope formula

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - (-1)}{-8 - (-3)}$$

$$= \frac{-4 + 1}{-8 + 3} = \frac{-5}{-5} = 1$$

OR $(-8, -4)$ and $(-3, -1)$
 (x_1, y_1) (x_2, y_2) $\rightarrow \frac{-5}{-5} = 1$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1 - (-4)}{-3 - (-8)} = \frac{-1 + 4}{-3 + 8}$$

Linear Function

EXAMPLE

03

Find an equation of the linear function given

$$f(2) = 5 \text{ and } f(6) = 3.$$

$$\Rightarrow f(2) = 5 \Rightarrow (2, 5)$$

$$f(6) = 3 \Rightarrow (6, 3)$$

$$b) m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 5}{6 - 2} = \frac{-2}{4} = -\frac{1}{2}$$

c. substitute the value of m in $y = mx + b$

$$5 = -\frac{1}{2} \cdot 2 + b$$

$$b = 5 + 1 = 6 \rightarrow y - \text{intercept}$$

812 (4,3)

$$y = mx + b$$

$$3 = -\frac{1}{2} \cdot 4 + b$$

$$b = 3 + 2 = 5 //$$

$$y = -\frac{1}{2}x + 5$$

$$f(x) = -\frac{1}{2}x + 5 //$$

Linear Function

EXAMPLE

- 04 Find an equation of the line passing the two given points:
 $(-5, -3)$ and $(-1, 2)$ and then graph the line.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - (-3)}{-1 - (-5)} = \frac{2 + 3}{-1 + 5} = \frac{5}{4}$$

$$y = mx + b$$
$$y = \frac{5}{4}x + b$$

① $(-5, -3)$

$$-3 = \frac{5}{4}(-5) + b$$
$$b = -3 + \frac{25}{4} = \frac{-12 + 25}{4} = \frac{13}{4}$$

② $(-1, 2)$

$$2 = \frac{5}{4}(-1) + b$$
$$b = 2 + \frac{5}{4} = \frac{8 + 5}{4} = \frac{13}{4}$$

Therefore :

$$\therefore y = \frac{5}{4}x + \frac{13}{4}$$

Linear Function

EXAMPLE

05

Find the coordinates of the point of intersection of the two given lines by drawing their graphs.

$$2x - y = 1 \text{ and } x + 2y = -7$$

$$\textcircled{1} 2x - y = 1$$

$$y = 2x - 1$$

x	y
0	-1
1	1
2	3
3	5
-1	-3
-2	-5

$$\textcircled{2} x + 2y = -7$$

$$2y = -x - 7$$

$$y = \frac{-x - 7}{2}$$

x	y
0	-7/2
1	-4
2	-9/2
-1	-3

At pt of intersection

